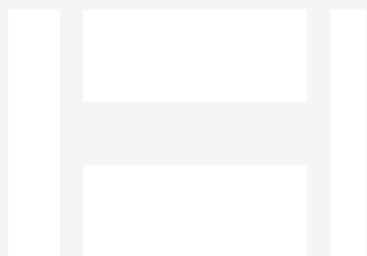


Hwacheon Mega 95 for Industrial & General Manufacturing

Long-bed CNC lathe for extended shafts, rollers, rods, and structural round parts. This paper covers what the machine does, the industrial & general manufacturing parts we produce with it, the alloys we run, the tolerances and finishes industrial & general manufacturing buyers expect, and the documentation package.



HWACHEON

Rattle & Hum · 16" OD × 120" L max turning · long-bed heavy-duty CNC lathe

Executive summary

Machine: Hwacheon Mega 95 (shop nickname: *Rattle & Hum*). long-bed heavy-duty CNC lathe at B&R Productions, running from our New Waverly, Texas shop under an ISO 9001:2015 quality system.

Application: Long-bed CNC lathe for extended shafts, rollers, rods, and structural round parts for Industrial & General Manufacturing customers — Industrial equipment OEMs, custom machinery builders, pump and valve manufacturers outside oilfield, food and pharma equipment suppliers, mining and heavy-equipment operators, reverse-engineering customers, sustainment for out-of-production industrial equipment.

Envelope: 16" OD × 120" L max turning. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

Ten feet of turning length is a specialty capability. Long drive shafts, extended mandrels, rig-floor rollers, oil-and-gas structural rods — the parts that don't fit on a 4- or 6-foot lathe. Rattle & Hum lets us keep this class of work in-house instead of outsourcing the finish turning after roughing on a larger machine somewhere else.

Full spec table

Model	Hwacheon Mega 95 ("Rattle & Hum")
Type	Heavy-duty gap-bed CNC lathe — Mega-95 platform
Max turning capacity	16" OD × 120" L (B&R working spec)
Platform max swing over bed	38.5"
Swing through gap	48.4"
Swing over cross slide	26.2"
Distance between centers	10 ft (3,000 mm) B&R config; platform available to 205"
X-axis travel	19.09"
Spindle bore	6"
Spindle speed	40 – 1,000 RPM (heavy-duty range)
Spindle motor	30 HP
Chuck	24" 3-jaw hydraulic
Turret	12-position
Control	Fanuc 21-T (typical)

Machine weight	~22,000 lb
Tolerance capability	±0.0005" routine on critical features; straightness held over full length
Materials	Full oilfield alloy range — Inconel, Duplex, 17-4 PH, 4140/4340, Monel

Who this serves in Industrial & General Manufacturing

Typical buyer: Industrial equipment OEMs, custom machinery builders, pump and valve manufacturers outside oilfield, food and pharma equipment suppliers, mining and heavy-equipment operators, reverse-engineering customers, sustainment for out-of-production industrial equipment.

What's hard about industrial & general manufacturing work: Legacy parts with no OEM support, one-off replacement components, small-lot production runs, materials that don't fit standard shop capabilities, and tight lead times on repair work where a down-machine is losing money by the day.

Typical Industrial & General Manufacturing parts we produce on Rattle & Hum

- Long industrial drive shafts (up to 10 feet)
- Extended pump shafts and spindles
- Rollers and extended rollers
- Any industrial turned part that busts 4-foot class
- Long structural rods and threaded parts

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Industrial & General Manufacturing

Standard range: Stainless (304, 316, 17-4 PH) · Tool steels · 4140 / 4340 · Brass · Bronze · Aluminum · Cast iron · Specialty alloys per customer requirement.

Industrial work covers the widest alloy range because the applications are the widest. Food and pharma often specify 316L or 17-4 PH; mining and heavy equipment specify tool steels and 4140/4340; pump manufacturers outside oilfield use everything from bronze bushings to stainless impellers. We work with the customer's spec sheet — if you tell us the alloy, we'll tell you honestly whether we can machine it and how quickly.

What "we run these weekly" means: we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

Tolerance and finish expectations in Industrial & General Manufacturing

±0.0005" routine on critical features. Industrial applications range from precision (pump seal areas, close-fit bushings) to structural (mining and heavy-equipment brackets); we adjust the inspection plan to what the part actually needs. Surface finish measured on parts where finish drives function.

Documentation and quality workflow

Documentation calibrated to customer requirement. Material traceability standard for critical parts. Certificates of conformance issued with delivery. NDAs on request for proprietary designs or reverse-engineered legacy parts.

Response time and emergency work

Down-machine and production-critical work prioritized. Lead times honestly quoted — no false promises. For legacy reverse-engineering, we typically confirm feasibility on the first phone call. Call (936) 291-7827.

How we compare

Compared to sourcing a new OEM part: we're often faster and cheaper for one-offs and small runs, and for legacy parts where the OEM has moved on we're often the only realistic option. Compared to a cheapest-bidder job shop: we hold tolerance, document material, and don't ruin the part.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request. ITAR-controlled prints handled under document control with prints staying in-shop.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Ready to talk about your Industrial & General Manufacturing work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

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